

Questões múltiplas

4.3 / 4.4

1 E

2

$$Q = 10^{10} \cdot e = 10^{10} \cdot 1,6 \times 10^{-19}$$

$$C = \frac{Q}{\Delta V} = \frac{10^{10} \cdot 1,6 \times 10^{-19}}{9,0} = 0,178 \times 10^{-9} = 178 \text{ pF}$$

C //

3

$$r = 5 \text{ cm}$$

$$d = 1 \text{ cm}$$

$$C_{\text{plano}} = \frac{K \cdot A}{4 \cdot \pi \cdot k \cdot d} = \frac{1 \cdot \pi \cdot 0,05^2}{4 \cdot \pi \cdot 9 \times 10^9 \cdot 0,01} = 6,94 \text{ pF}$$

A //

4

$$\left\{ \begin{array}{l} U = \frac{1}{2} \cdot Q \cdot \Delta V \\ \Delta V = Q / C \\ C_{\text{plano}} = \frac{K \cdot A}{4 \cdot \pi \cdot k \cdot d} \end{array} \right\} \Rightarrow U = \frac{1}{2} \cdot \frac{Q^2}{C} \Rightarrow U = \frac{1}{2} \cdot Q^2 \cdot \frac{4 \cdot \pi \cdot k \cdot d}{K \cdot A}$$

$$U_1 = \frac{(3 \mu)^2 \cdot 2 \cdot \pi \cdot k \cdot 0,001}{K \cdot A}$$

$$\frac{U_2}{U_1} = \frac{(9 \mu)^2 \cdot 0,003}{(3 \mu)^2 \cdot 0,001} = 27$$

$$U_2 = \frac{(9 \mu)^2 \cdot 2 \cdot \pi \cdot k \cdot 0,003}{K \cdot A}$$

D //

5

C //

Problemas

1)

$$\begin{aligned}P &= 2 \text{ kW} \\ \Delta t &= 2 \text{ ms} \\ C &= 50 \mu\text{F}\end{aligned}$$

a)

$$\left| \begin{aligned}U &= \frac{1}{2} \cdot Q \cdot \Delta V \\ C &= \frac{Q}{\Delta V}\end{aligned} \right| \Leftrightarrow \left| \begin{aligned}U &= \frac{1}{2} \cdot C \cdot \Delta V^2 \Leftrightarrow \Delta V = \sqrt{\frac{2 \cdot U}{C}}\end{aligned} \right|$$

$$U = P \cdot \Delta t = 2 \text{ k} \cdot 2 \text{ m} = 4 \text{ J}$$

$$\Delta V = \sqrt{\frac{2 \times 4}{50 \mu}} = 400 \text{ V}$$

b)

$$C = 250 \mu\text{F}$$

$$\Delta V = \sqrt{\frac{2 \times 4}{250 \mu\text{F}}} = 779 \text{ V}$$

c)

Capa um maior volume.

2)

a)

$$r = 4,0 \text{ cm}$$

$$C_{\text{esf. isol.}} = \frac{K \cdot R}{k} = \frac{1 \times 0,04}{9 \times 10^9} = 4,44 \text{ pF}$$

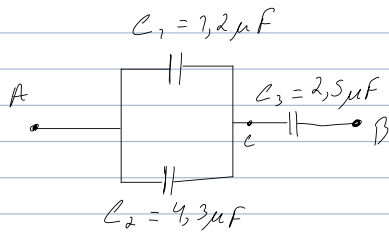
b)

$$C_{\text{esf}} = \frac{K \cdot R_1 \cdot R_2}{k \cdot (R_2 - R_1)} = \frac{5,6 \times 0,04 \times 0,041}{9 \times 10^9 \cdot (0,041 - 0,04)} = 1,02 \text{ nF}$$

c)

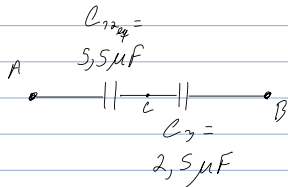
$$\frac{C_{\text{esf}}}{C_{\text{esf. isol.}}} = \frac{1,02 \text{ n}}{4,44 \text{ p}} = 230 \text{ F/F}$$

3



$$\Delta V_{AB} = 9,0 \text{ V}$$

a)



$$C_{\text{Total}} = \frac{5,5 \mu \cdot 2,5 \mu}{5,5 \mu + 2,5 \mu} = 1,72 \mu \text{ F}$$

$$Q_{\text{Total}} = C_{\text{Total}} \cdot \Delta V = 1,72 \mu \times 9,0 = 15,5 \mu \text{ C} = Q_{12eq} = Q_3$$

$$\Delta V_{AC} = \frac{Q_{12eq}}{C_{12eq}} = \frac{15,5 \mu}{5,5 \mu} = 2,82 \text{ V}$$

$$Q_1 = \Delta V_{AC} \cdot C_1 = 2,82 \cdot 1,2 \mu = 3,38 \mu \text{ C}$$

$$Q_2 = \Delta V_{AC} \cdot C_2 = 2,82 \cdot 4,3 \mu = 12,1 \mu \text{ C}$$

b)

$$\begin{cases} U = \frac{1}{2} \cdot Q \cdot \Delta V \\ Q = C \cdot \Delta V \end{cases} \Rightarrow U = \frac{1}{2} \cdot C \cdot \Delta V^2 = \frac{1}{2} \cdot 1,72 \mu \cdot 9^2 = 69,7 \mu \text{ J}$$

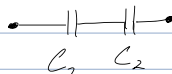
4

Condensador plano com dois dielétricos com espessura igual a 9,5 cm
 $A = 12 \text{ cm}^2$ em que $K_1 = 4,9$ e $K_2 = 5,6$
 $d = 1 \text{ cm}$

Sugestão: Pensar em dois condensadores em série com dielétricos diferentes.

$$C = ?$$

$$C_{\text{plano}} = \frac{K \cdot A}{4 \cdot \pi \cdot k \cdot d}$$



$$C_1 = \frac{4,9 \cdot 12 \times 10^{-4}}{4 \cdot \pi \cdot 9 \times 10^9 \cdot 0,005} = 10,4 \text{ pF}$$

$$C_2 = \frac{5,6 \cdot 12 \times 10^{-4}}{4 \cdot \pi \cdot 9 \times 10^9 \cdot 0,005} = 11,9 \text{ pF}$$

$$C_{eq} = \frac{C_1 \cdot C_2}{C_1 + C_2} = \frac{10,4 \text{ p} \cdot 11,9 \text{ p}}{10,4 \text{ p} + 11,9 \text{ p}} = 5,55 \text{ pF}$$

5

Condensador plano; $A = 0,3 \text{ m}^2$; $d = 0,5 \text{ cm}$; archedo como dielétrico; $\Delta V = 12 \text{ V}$

a)

$$C_{\text{plano}} = \frac{K \cdot A}{4\pi \cdot k \cdot d} = \frac{3,4 \times 0,3}{4 \cdot \pi \cdot 9 \times 10^9 \cdot 0,005} = 1,80 \text{ mF}$$

$$\begin{cases} U = \frac{1}{2} \cdot Q \cdot \Delta V \\ Q = C \cdot \Delta V \end{cases} \Rightarrow U = \frac{1}{2} \cdot C \cdot \Delta V^2 = \frac{1}{2} \cdot 1,80 \text{ m} \cdot 12^2 = 129,6 \text{ mJ}$$

(...)

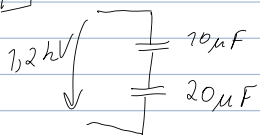
b)

$$V_{\text{max}} = E_{\text{máx}} \cdot d$$

$$V_{\text{máx archedo}} = 40 \overset{\text{kV/mm}}{\times} 5 = 200 \text{ kV}$$

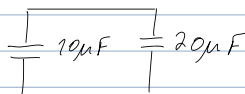
$$V_{\text{máx ar}} = 3 \times 5 = 15 \text{ kV}$$

6



$$a) Q_1 = Q_2 = Q_T = C \cdot \Delta V = \left(\frac{10\mu \cdot 20\mu}{10\mu + 20\mu} \right) \cdot 1200 = 8 \text{ mC}$$

b)



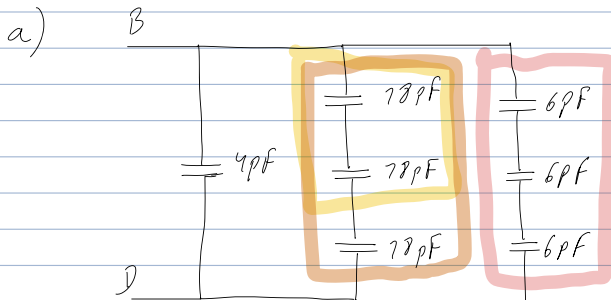
$$Q_T = 2 \times 8 \text{ mC} = 16 \text{ mC}$$

$$\Delta V = \frac{Q_T}{C_{eq}} = \frac{16 \text{ m}}{10 \mu + 20 \mu} = 533 \text{ V}$$

$$Q_1 = C_1 \cdot \Delta V = 10 \mu \cdot 533 = 5,33 \text{ mC}$$

$$Q_2 = C_2 \cdot \Delta V = 20 \mu \cdot 533 = 10,66 \text{ mC}$$

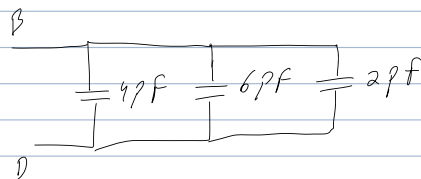
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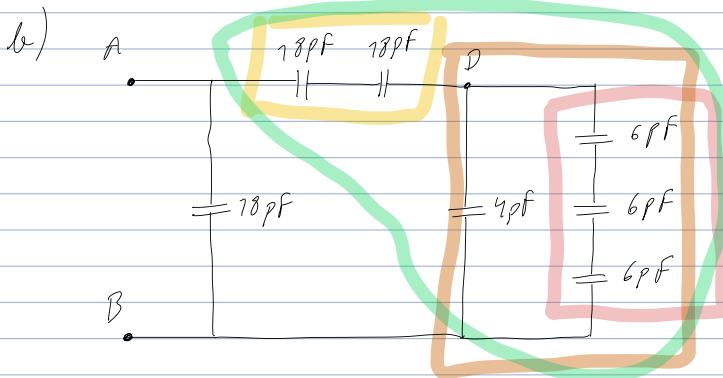
$$C_1 = \frac{18p \cdot 18p}{18p + 18p} = 9pF$$

$$C_2 = \frac{9p \cdot 18p}{9p + 18p} = 6pF$$

$$C_3 = (\dots) = \frac{1}{3} \cdot 6p = 2pF$$



$$C_T = 4p + 6p + 2p = 12pF$$



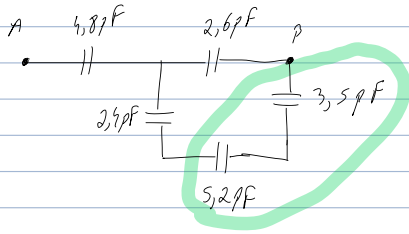
$$C_1 = (\dots) = \frac{1}{3} \cdot 6p = 2pF$$

$$C_2 = (\dots) = \frac{1}{2} \cdot 18p = 9pF$$

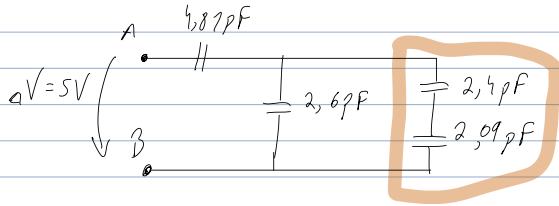
$$C_3 = 4p + 2p = 6pF$$

$$C_4 = \frac{9p \cdot 6p}{9p + 6p} = 3,6pF$$

$$C_T = 18p + 3,6p = 21,6pF$$

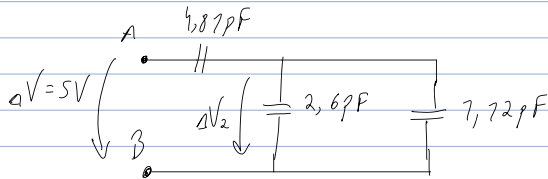


$$C_1 = \frac{3,5p \cdot 5,2p}{3,5p + 5,2p} = 2,09 pF$$



$$Q_{2,4p} = Q_{2,09} = Q_{2,4 \parallel 2,09}$$

$$C_2 = \frac{2,4p \cdot 2,09p}{2,4p + 2,09p} = 1,12 pF$$



$$C_{eq} = \frac{4,87p \cdot (2,6p + 7,72p)}{4,87p + (2,6p + 7,72p)} = 2,70 pF$$

$$Q_{eq} = C_{eq} \cdot \Delta V = 2,70p \cdot 5 = 10,50 pC = Q_{4,87p} = Q_{(2,6p + 7,72p)}$$

$$\Delta V_2 = \frac{Q_{(2,6p + 7,72p)}}{(2,6p + 7,72p)} = 2,82 V$$

$$Q_{7,72p} = 7,72p \cdot 2,82 = 3,76 pC = Q_{2,4p}$$